

המשובצת

Full Stack Final Project
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THE PROBLEM

- Staffing many shifts with different requirements
- Choosing the correct people out of a large pool
- Ensuring the best possible combinations
- Ensuring minimum mistakes

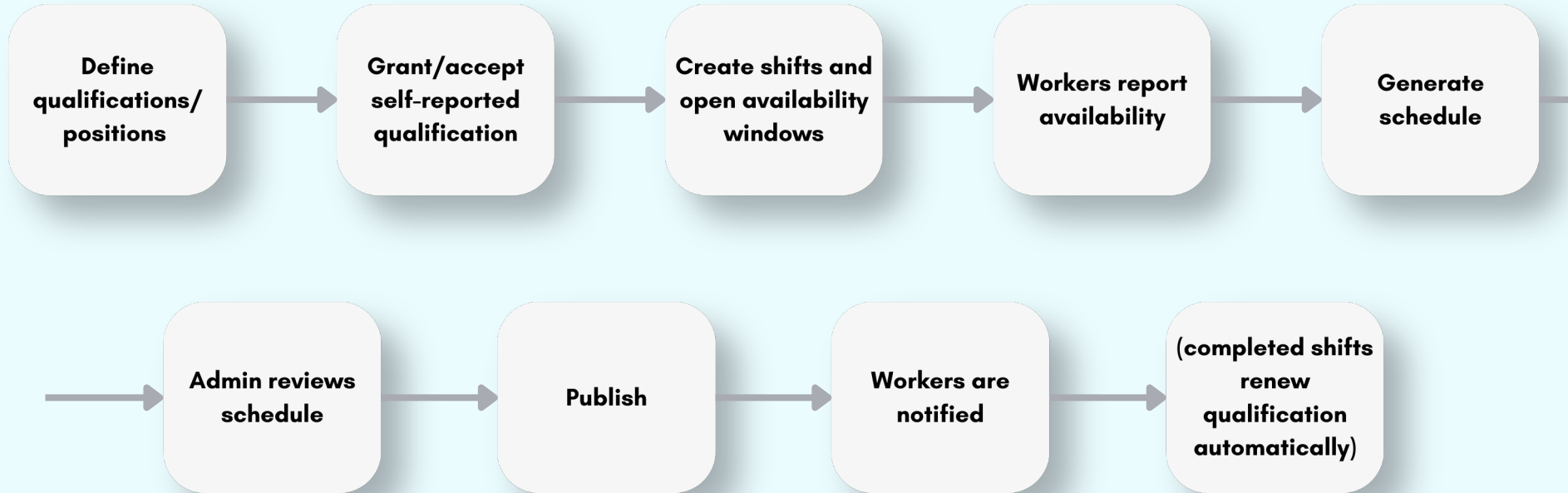
USERS & BUSINESS VALUE

Admin: defines positions/qualifications, manages schedules

Worker: views qualification/shifts, submits availability

Value: less time building the schedule, fewer errors, qualification tracking over time

CORE PRODUCT FLOW



DEMO

<https://full-stack-project-neon-pi.vercel.app/>

ARCHITECTURE

Next.js:

**Backend =
Server components
+ Server actions
(No separate API)**

Supabase:

**Postgres +
Authentication +
Row-Level security**

Vercel hosting:

**Hosts and deploys
the website**

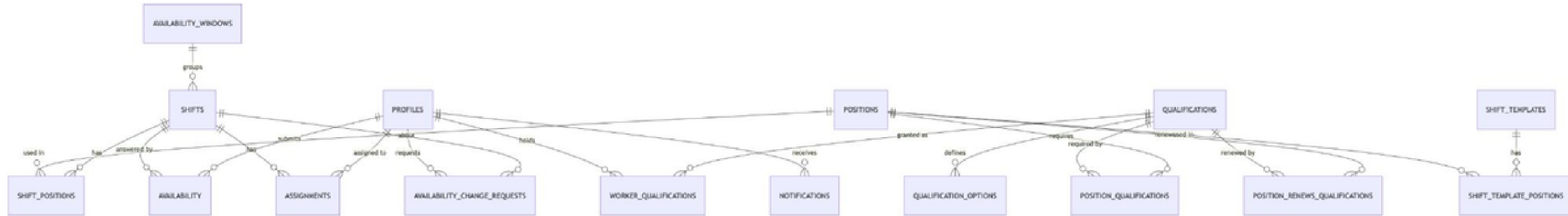
DATABASE

Shifts

Profiles

Positions

Qualifications



THE SCHEDULING ENGINE

Greedy heuristic, not a full optimizer

1 Flatten the
slots

2 Check
eligibility

3 Sort by
scarcity

4 Take
constraints into
consideration

5 Assign

6 Flag gaps

***Admin reviews before publishing**

TESTING

- **4 Layers: unit, component, integration (real Supabase + real RLS), documented manual**
- **68 automated tests total**
- **No playwright - substituted with a real-Supabase integration layer instead**

SCALING

- **Real ceiling: a squadron roster - tens to couple hundred people**
- **Scaled good: indexes, connection pooling, the engine's runtime**
- **Known limitation: no pagination yet, one unbounded query ('getLatestRenewalDates')**

SECURITY

- **Two independent layers:**
 - **app-level roleguard**
 - **Postgres Row-Level Security (the real boundary)**
- **Secrets:**
 - **Anonkey is meant to be public**
 - **Service role key is server only**

WHAT I'D IMPROVE WITH MORE TIME

- **Pagination for shift history / notifications**
- **Add a 'workers with qualifications that are about to expire' preference to the engine**
- **Real email invites and email confirmation**
- **Targeted "shift needs volunteers" notifications instead of only flagging understaffed shifts**

THANK YOU!